

CS-456 Computer Vision — Complex Computing Problem

Autonomous Visual Intelligence System for Scene Understanding

Technical Report | CLOs 3, 4, 5 | classical + deep-learning pipeline

1. Executive Summary

This project implements the four core perception capabilities an outdoor navigation robot needs, each built and independently validated on a representative real, publicly licensed dataset (not a single shared feed). Task 1 detects road edges and lane lines (Gaussian -> Canny -> probabilistic Hough) on real dashcam road images. Task 2 establishes cross-view correspondence (SIFT + Lowe's ratio test) between two real views of the same outdoor landmark. Task 3 builds a real panorama from overlapping building frames using a from-scratch RANSAC homography estimator and multi-band blending on real overlapping outdoor waterfront frames (no cv2.Stitcher). Task 4 fine-tunes the classification head of a frozen ResNet-18 to label four outdoor scene categories (road, building, vegetation, pedestrian zone), reaching 93.8% validation accuracy (target 70%). Task 5 quantifies and critiques the accuracy, robustness and trade-offs of each stage.

2. Pipeline Architecture

Autonomous navigation robot — perception stack

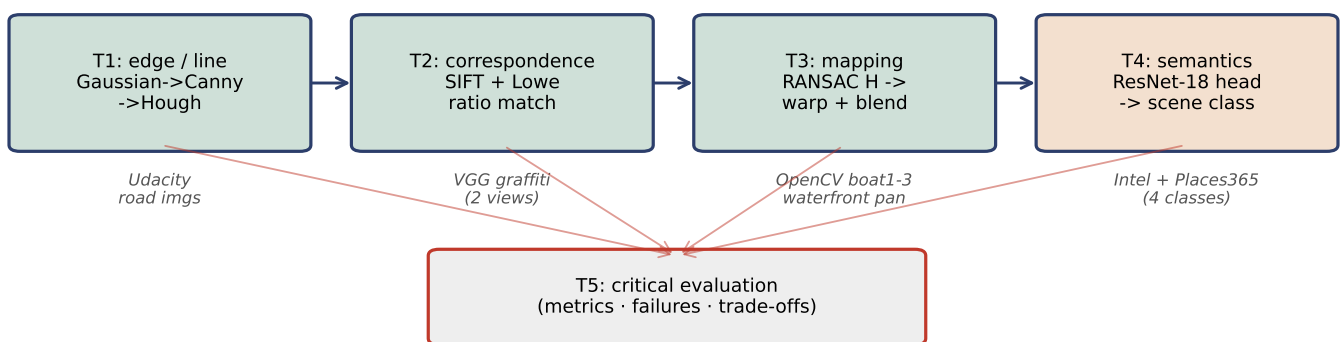


Figure 1 — The four perception capabilities of the robot's vision stack. Each is implemented and validated independently on the real dataset named beneath it (italic); Task 5 evaluates all four. They are not run on one shared frame.

Headline metrics: T1 edge density (balanced) = 1.11%, 86 fine-Hough lines | T2 keypoints A/B = 2674/3506, good = 539 | T3 mean RANSAC inlier ratio = 86.0% | T4 best val acc = 93.8%

3. Quantitative Results

Per-task metrics (from outputs/results.json):

T1 edge-detection quality: balanced Canny edge density = 1.11% of pixels; 86 lines at the 'fine' Hough setting.

T2 match ratio: 539 good matches / 2674 vs 3506 keypoints (ratio 0.202).

T3 RANSAC inlier ratio: mean 86.0% (multiband blend, ref frame 2).

T4 CNN accuracy: best 93.8% / final 93.8% validation.

T1 — Canny experiments (with qualitative outcomes)

Task 1 — Canny threshold experiments

image	config	low/high	edge density	qualitative outcome
straight_road	loose	30/90	2.89%	many weak edges; foliage/road texture clutter
straight_road	balanced	50/150	0.95%	clean lane + structure edges (1:3 ratio)
straight_road	strict	100/200	0.60%	only strong edges; faint/lane lanes dropped
curved_road	loose	30/90	2.09%	many weak edges; foliage/road texture clutter
curved_road	balanced	50/150	1.25%	clean lane + structure edges (1:3 ratio)
curved_road	strict	100/200	0.52%	only strong edges; faint/lane lanes dropped
lane_switch	loose	30/90	1.92%	many weak edges; foliage/road texture clutter
lane_switch	balanced	50/150	1.14%	clean lane + structure edges (1:3 ratio)
lane_switch	strict	100/200	0.31%	only strong edges; faint/lane lanes dropped

T2 — SIFT matching statistics

Task 2 — SIFT matching statistics

metric	value
keypoints (A)	2674
keypoints (B)	3506
raw knn candidates	2674
good matches (ratio 0.75)	539
top matches drawn	50
match quality ratio	0.202

T4 — classifier summary

Task 4 — Summary

metric	value
backbone	resnet18
classes	building, pedestrian_zone, road, vegetation
images/class (train)	160
epochs	8
final val accuracy	0.938
best val accuracy	0.938
target (70%) met?	YES
Grad-CAM samples	3

T3 — pairwise RANSAC inlier ratios

Task 3 — pairwise stitching stats

pair	#matches	RANSAC inliers	note
1-2	927	89.5% (830/927)	ok
2-3	766	82.5% (632/766)	ok

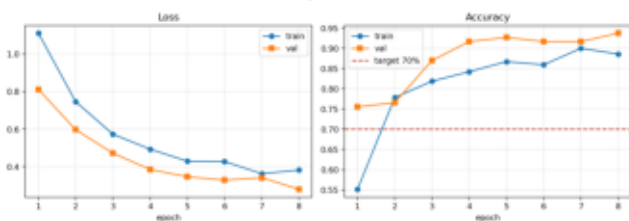
T4 — validation confusion matrix

Task 4 — Confusion matrix (validation)

	building	pedestrian_zone	road	vegetation
building	46	0	2	0
pedestrian_zone	0	47	1	0
road	6	2	40	0
vegetation	1	0	0	47

T4 — training / validation curves

Task 4 — Training / validation curves



T4 — per-class precision / recall / F1

Task 4 — Per-class precision / recall / F1

class	precision	recall	F1	support
building	0.87	0.96	0.91	48
pedestrian_zone	0.98	0.98	0.97	48
road	0.93	0.83	0.88	48
vegetation	1.00	0.98	0.99	48

4. Failure Analysis (≥ 2 modes per task, with visuals)

Task 1 - Edge & line detection.

(a) Loose Canny thresholds (30/90) flood the map with road-texture/foilage clutter: edge density ranges ~ 0.3 - 2.1% across configs, so Hough returns many spurious short segments. (b) Strict thresholds (100/200) instead drop faint far-field lane markings. Mitigation: the 1:3 'balanced' threshold + a moderate maxLineGap.

Task 2 - SIFT matching.

(a) A high ratio (0.9) admits many false positives (1212 matches) while a low ratio (0.5) starves recall (87 matches) - the sweep quantifies the precision/recall trade-off. (b) Repetitive wall texture yields ambiguous descriptors and geometrically inconsistent matches.

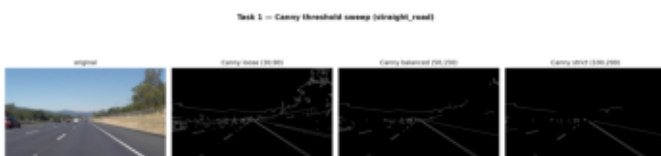
Task 3 - Panorama.

(a) Insufficient overlap ($< \text{min matches}$) makes the homography rank-deficient - detected and reported as an edge case. (b) Accumulated homography error across frames bends lines near the panorama edges (mean inlier ratio 86.0%); middle-frame anchoring limits this drift.

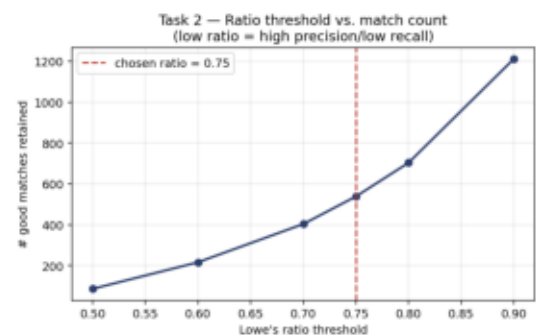
Task 4 - Scene classification.

(a) Visually similar classes are confused (most confused: road $\rightarrow$ building, 6 imgs) - see the confusion-matrix off-diagonals. (b) The Grad-CAM 'incorrect' example shows the network attending to background context rather than the discriminative structure, explaining the misclassification.

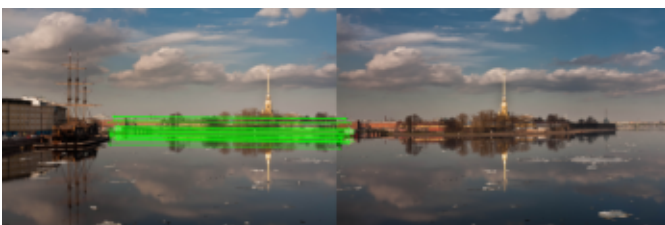
T1 - Canny threshold sweep



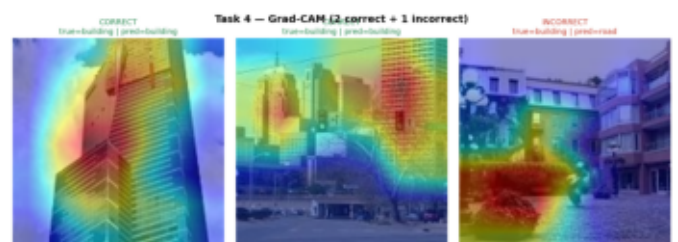
T2 - matches vs Lowe ratio



T3 - RANSAC inlier matches (pair 1)



T4 - Grad-CAM (2 correct + 1 incorrect)



5. Classical vs. Deep-Learning Trade-offs

Dimension	Classical (T1-T3)	Deep learning (T4)
Speed	Real-time on CPU	Needs GPU/MPS to train; fast at inference
Accuracy	Exact when geometry holds	High on semantics, degrades off-distribution
Interpretability	Fully transparent maths	Opaque; needs Grad-CAM to explain
Data requirement	None (hand-tuned params)	≥ 100 labelled images/class
Failure mode	Breaks on low texture/overlap	Confuses visually similar classes

6. Limitations & Future Work

- Each capability is validated on its own public dataset; a true robot would run all four on one live camera stream, adding motion blur, exposure change and temporal consistency needs.
- Task 4 is a frozen-backbone linear probe on four classes; unfreezing deeper blocks and scaling to the full Places365 taxonomy would test generalisation and long-tail robustness.
- Homographies assume a largely planar / rotational scene; strong parallax from a moving ground robot would need bundle adjustment or cylindrical warping.
- SIFT + FLANN are CPU-bound; a learned detector (SuperPoint) or GPU matcher would raise throughput.
- Future work: fuse the geometric map (T1-T3) with the semantic labels (T4) into a temporal scene graph.

7. Data Sources & Citations

- T1 roads — Udacity CarND-LaneLines-P1 test images (MIT License).
- T2 pair — Oxford VGG affine 'graffiti' wall, views 1 & 3 (free for research; via opencv_extra).
- T3 frames — OpenCV stitching sample 'boat1-3', a St. Petersburg waterfront panorama (opencv_extra, BSD-3).
- T4 road/building/vegetation — Intel Image Classification photos (Kaggle).
- T4 pedestrian_zone — Places365 'crosswalk' (Zhou et al., IEEE TPAMI 2017).
- No images are self-captured; all inputs are real, publicly-licensed datasets.

Reproducibility: fixed global seed; every figure regenerated by 'python run_all.py'.